

Description-aware Coarse-to-Fine Fashion Image Inpainting

Furkan Kinli

Department of Computer Science

Özyeğin University

furkan.kinli@ozyegin.edu.tr

Abstract

Inpainting a particular missing region in an image is a challenging vision task, and promising improvements on this task have been achieved with the help of the recent developments in vision-related deep learning approaches. Although it may have a direct influence on the decisions of fashion analysis systems, a limited number of studies for image inpainting task have been done in fashion domain. In this study, we propose a multi-modal generative learning approach for filling the missing parts in fashion images by using their descriptions besides to the visual features. Our model is composed of four main blocks which are textual feature extractor, coarse image generator guided by textual features, fine image generator enhancing the coarse output, and lastly global and local discriminators improving both coarse and fine outputs. Experiments on FashionGen dataset show that our multi-model approach is able to generate visually plausible patch structures and textures for the missing parts in the images.

Dataset



Figure 1. Example images and their descriptions from FashionGen dataset.

Methodology



Figure 2. The overall architecture design of our network.

Results

- Transpose convolution over Bilinear Up-sampling
- Instance Normalization over Batch Normalization
- Content loss increases the overall complexity too much, but changes the results insignificantly
- Constraints by textual features leads to faster convergence

